

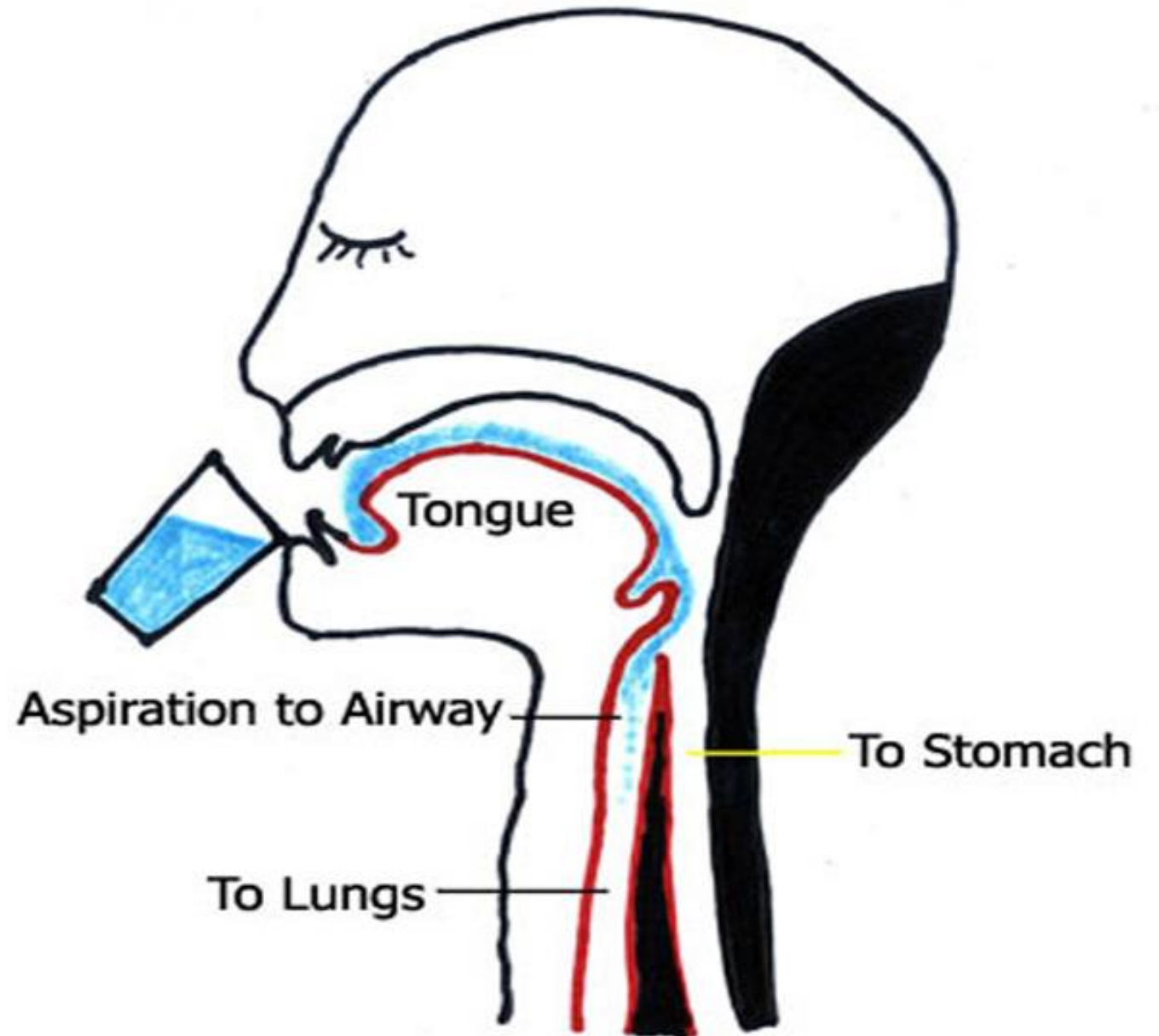


Physiology of Swallowing

Dr. Nazleen Shakir Akreyi

GIT Module

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Learning outcomes

- Understand the physiology of swallowing
- Describe the three phases of deglutition

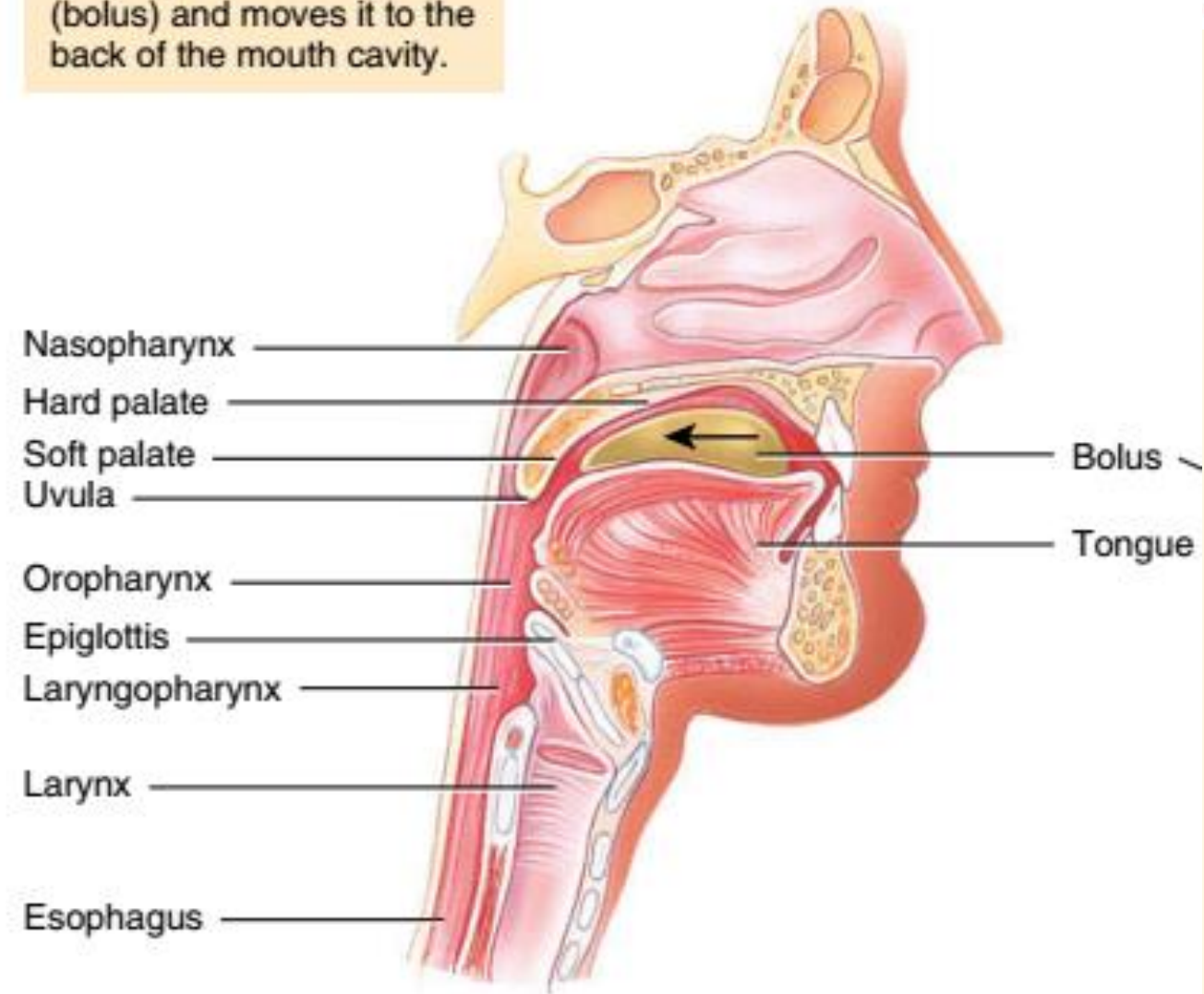
Swallowing (Deglutition)

- Is the movement of food from the mouth into the stomach
- It is facilitated by the secretion of saliva and mucus
- It involves the mouth, pharynx, and esophagus
- It occurs in three steps:
 1. The voluntary stage
 2. The pharyngeal stage
 3. The esophageal stage

1. The voluntary stage

- Swallowing starts when the bolus is forced to the back of the oral cavity and into the oropharynx by the movement of the tongue upward and backward against the palate

The tongue shapes the chewed, lubricated food (bolus) and moves it to the back of the mouth cavity.



(a) Position of structures during voluntary stage

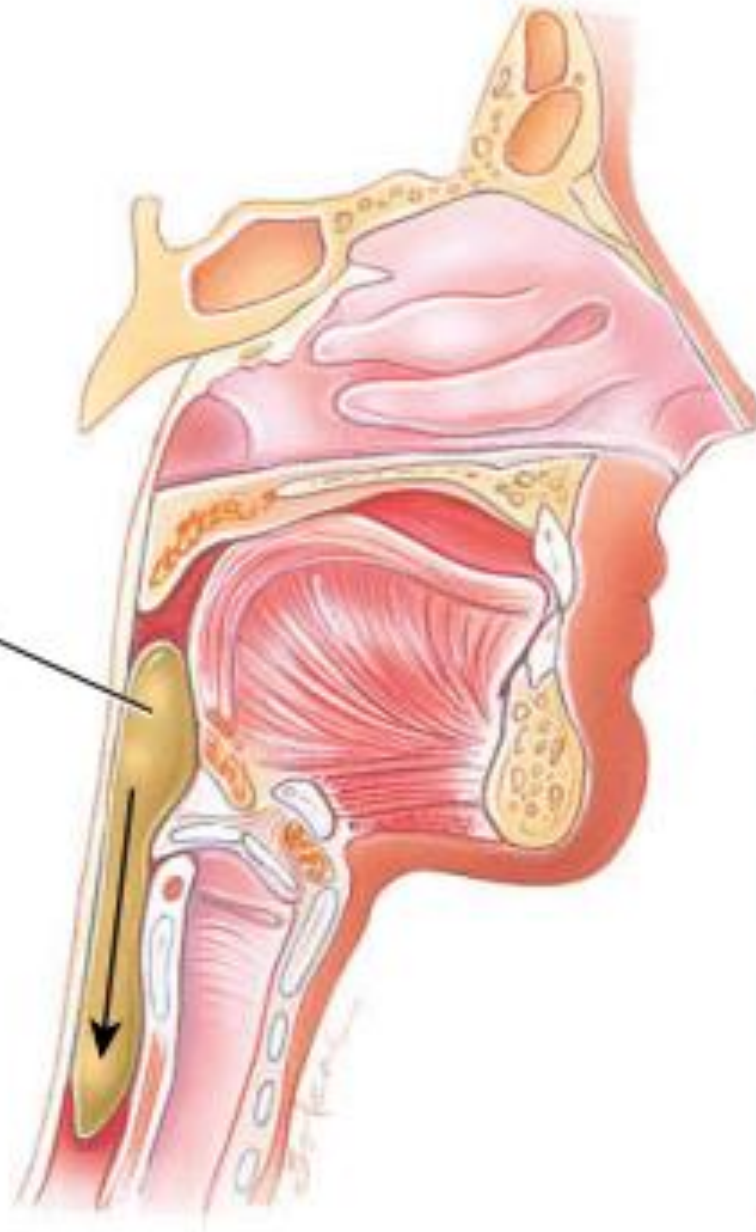
2. The pharyngeal stage

- This phase is **involuntary**
- It begins when the bolus stimulates receptors in the oropharynx, which send impulses to the **deglutition center** in the medulla oblongata and lower pons of the brain stem
- The returning impulses cause the **soft palate** and **uvula** to move **upward** to close off the nasopharynx to prevent swallowed foods and liquids from entering the nasal cavity

2. The pharyngeal stage. Cont.

- The **epiglottis closes** off the opening to the larynx, which prevents the bolus from entering the rest of the respiratory tract
- The bolus moves through the oropharynx and the laryngopharynx
- Once the upper esophageal sphincter relaxes, the bolus moves into the esophagus

Bolus



- The tongue rises against the palate and closes the nasopharynx.
 - The uvula and palate seal off the nasal cavity.
 - The epiglottis covers the larynx.
- Breathing is temporarily interrupted.

(b) Pharyngeal stage of swallowing

3. The esophageal stage

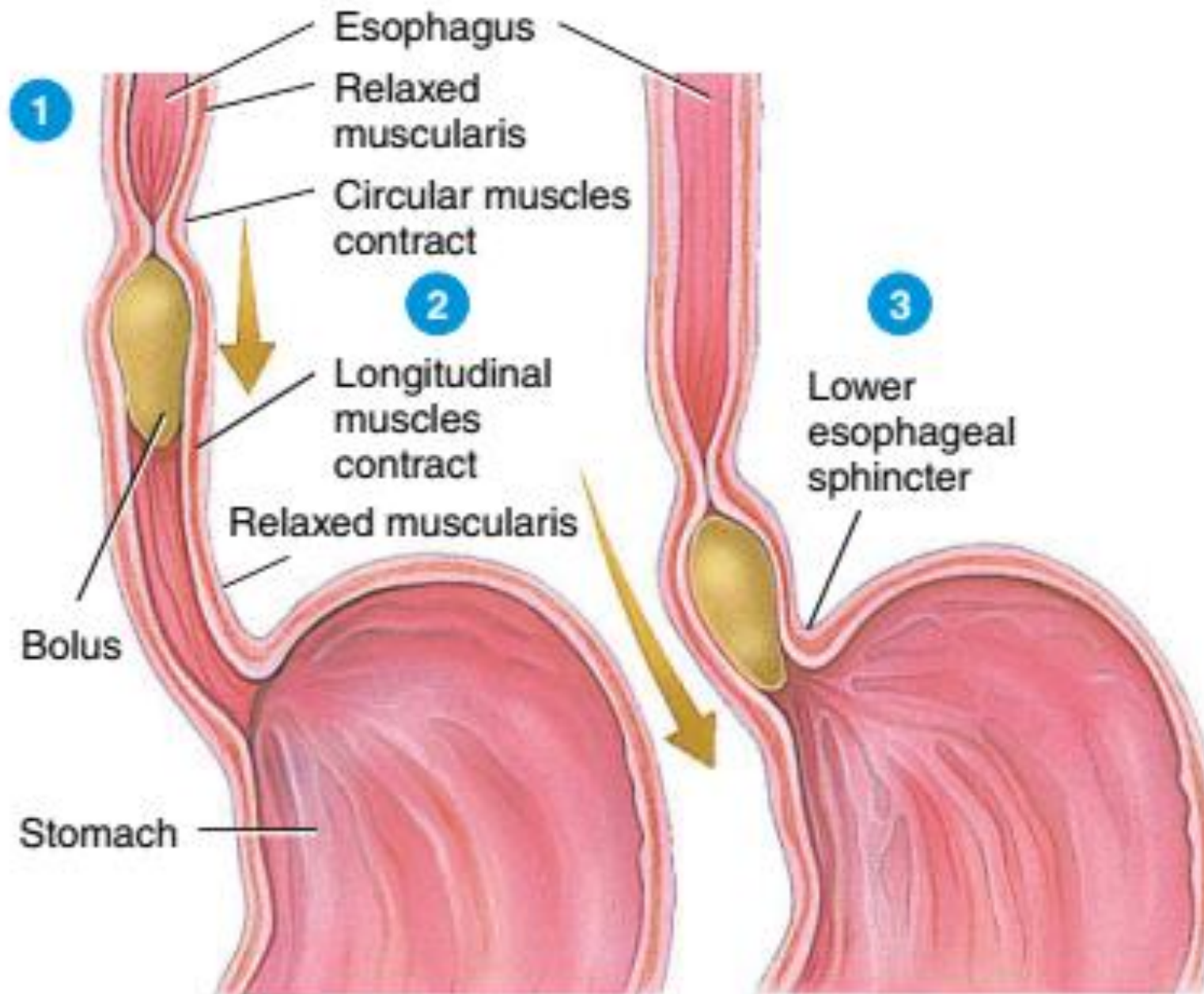
- It begins once the bolus enters the esophagus
 - During this phase, **peristalsis** (a progression of coordinated contractions and relaxations of the circular and longitudinal layers of the muscularis) pushes the bolus onward
- 1) In the section of the esophagus just **superior to the bolus**, the circular muscle fibers contract, constricting the esophageal wall and squeezing the bolus toward the stomach

3. The esophageal stage

2) Longitudinal fibers **inferior to the bolus** also contract, which shortens this inferior section and pushes its walls outward so it can receive the bolus. The contractions are repeated in waves that push the food toward the stomach

- Steps 1 and 2 are repeated until the bolus reaches the lower esophageal sphincter muscles

3) The lower esophageal sphincter relaxes, and the bolus enters the stomach



(c) Esophageal stage of swallowing

Swallowing

- Mucus secreted by esophageal glands lubricates the bolus and reduces friction
- The passage of solid or semisolid food from the mouth to the stomach takes 4 to 8 seconds
- Very soft foods and liquids pass through in about 1 second

